## Remarks on 'Lexical DP Blocking' in the Person-Case Constraint: Comparative evidence from San Juan Piñas Mixtec

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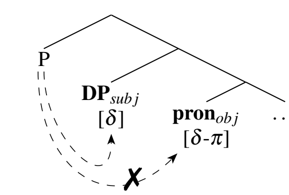
### 1 Introduction

Many languages display person restrictions on combinations of two pronominal clitics, a phenomenon now known as the Person-Case Constraint (PCC) (Perlmutter, 1971; Bonet, 1991, et seq.). A common approach to the PCC is that it arises from a single probe attempting but failing to Agree with both clitics, but individual analyses vary in the Agree mechanisms being posited and in the loci and feature specifications of the probe and goals.[1]

While most prior work has focused on internal arguments of ditransitives (though see Stegovec 2019; Deal 2024), Sichel and Toosarvandani (2024) show that, in Southeastern Sierra Zapotec ('SSZ'), an Eastern Otomanguean language group, the PCC restricts combinations of monotransitive external and internal arguments (henceforth, subjects and objects).[2] In addition to the cross-linguistically familiar restriction on pairs of clitics, the authors observe that SSZ also displays a different PCC-like pattern, otherwise unattested in the typology of hierarchy effects: object clitics of *all persons* (i.e., 1, 2, *and* 3) are banned when the subject is a lexical DP rather than a pronominal clitic, (1c).[3] Sichel and Toosarvandani frame the latter as a *Lexical DP Blocking effect (LDB)*. They propose that lexical DPs and pronouns share featural content (a D-feature, [$\delta$]), so both may in principle interact with an Agreeing probe P; due to this, coupled with their proposed conditions on Agree, a lexical DP subject systematically intervenes between a higher P and the object pronoun, preventing it from cliticizing to the verb, (2).[4]

(1) *Blenh$^{3}$=**eb$^{3}$**$_i$ Xwanh$^{1}$=a'$^{3}$ $t_i$
CMPL.carry=3.AN Juana=DEF
Intended: 'Juana carried it.'
(Laxopa; Sichel and Toosarvandani 2024)

(2)



The primary aim of this paper is to further elucidate the empirical and theoretical status of lexical DPs in hierarchy effects, by bringing forth novel data from the San Juan Piñas variety of Mixtec (SJPM), another Eastern Otomanguean language that we argue displays the PCC (restricting clitic combinations) and the LDB (restricting DP/clitic combinations) in monotransitives. The LDB's existence in Mixtec is typologically notable, suggesting that it may be a general property of Eastern Otomanguean languages. Crucially, however, the LDB in SJPM contrasts with that in SSZ in two important respects, (3): (i) certain DP/pronoun

[1] In addition to the analyses discussed in this paper, see Anagnostopoulou 2003; Béjar and Rezac 2003; Nevins 2011; Pancheva and Zubizarreta 2018, and Stegovec 2020. There are also approaches wherein both clitics are successfully targeted, with the PCC arising from overagreement (e.g. Coon and Keine, 2021).

[2] For reasons of space, this paper focuses on Sichel and Toosarvandani 2024, although related findings are reported in Toosarvandani 2017, Foley et al. 2019, Sichel and Toosarvandani 2020, Foley and Toosarvandani 2022, Toosarvandani 2023, and Foley et al. to appear. These authors consider the varieties spoken in the towns of Santiago Laxopa, Santa María Yalina, and San Sebastián Guiloxi, with data coming from their own fieldwork, as well as the varieties spoken in Hidalgo Yalálag (citing Avelino Becerra 2004 and López and Newberg 2005) and San Bartolomé Zoogocho (mostly building on Sonnenschein 2004). Data and ideas from such works that are not also repeated in Sichel and Toosarvandani 2024 will be discussed whenever relevant. As there are different data sources, orthographic conventions vary across the SSZ examples presented.

[3] Throughout this paper, tones (when given in the source examples) are encoded as superscript numerals, though the transcription conventions differ across languages.

[4] **Abbreviations for SSZ and SJPM data:** AN = animate, ARB = arboreal noun class, BASE = pronominal base, CL = classifier, CMPL = completive, CONT = continuative, DEF = definite, EL = elder, EX = exclusive, F = feminine, HAB = habitual, HON = honorific, HU = human, IRR = irrealis, M = masculine, N = neuter, NEG = negation, PL = plural, SG = singular, 1/2/3 = 1st/2nd/3rd person.

combinations ruled out in SSZ are well-formed in SJPM, and (ii) pronominal object clitics in SJPM visibly encliticize to DP subjects rather than raising past them.[5]

(3) a. *ko$^5$ni$^1$ [ ti$^5$ vi$^3$lu$^5$ ]=**$^n$di$^3$**
CONT.love CL.3.AN cat =1PL.EX
Intended: 'The cat loves us (excl.).'

b. ko$^5$ni$^1$ [ ti$^5$ vi$^3$lu$^5$ ]=**ɾa$^3$**
CONT.love CL.3.AN cat =3SG.M
'The cat loves him.' (SJPM)

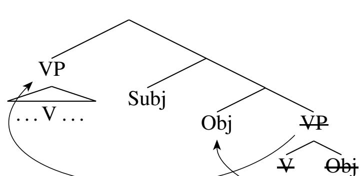
Assuming an otherwise uniform morphosyntax between the two languages (despite the different clitic positions), this paper evaluates whether Sichel and Toosarvandani's (2024) intervention-based analysis of the LDB can be extended to SJPM. It also considers Deal's (2024) recent Interaction/Satisfaction theory of the PCC, which the authors specifically name as unable to capture the LDB. We focus on these accounts because they differ not only in the theorization of Agree but also in the structural height of the probe needed to derive the empirical patterns. We show that both accounts can broadly accommodate the core LDB patterns in SSZ and SJPM (modulo certain stipulations)—but we also observe that ditransitives in both languages present an empirical challenge, in that they do not conform to the generalizations above and are difficult to integrate into either account. Although space considerations restrict us to comparing just these two analyses, we intend for this paper to serve as a blueprint for future work on this topic.

This paper is organized as follows. §2 presents the hierarchy effects found in SSZ monotransitives and summarizes the analysis of Sichel and Toosarvandani 2024. In §3 we turn to SJPM, and highlight the additional assumptions that Sichel and Toosarvandani's account would require to capture the similar yet non-identical SJPM facts. §4 then develops an alternative account building on Deal 2024, and explores its implications for nominal licensing and clause structure. Finally, in §5 we introduce recalcitrant data from ditransitives in both languages and highlight the challenges they pose for both analyses.

## 2 The PCC and LDB in Southeastern Sierra Zapotec

### *2.1 Overview of morphosyntactic patterns*

Zapotec is part of the Eastern Otomanguean subgroup of the larger Otomanguean language family (Kaufman, 1988; Campbell, 2017a,b). The varieties discussed by Sichel and Toosarvandani (2024) represent multiple Northern Zapotec varieties from the southeastern Sierra Norte region of Oaxaca, Mexico. The basic facts presented in this section hold for all varieties surveyed by these authors (collectively grouped under Southeastern Sierra Zapotec, or 'SSZ'). SSZ has a base V(erb)-S(ubject)-O(bject) word order. Following Adler et al. 2018, Sichel and Toosarvandani assume that verb-initial order in these languages is derived by object shift out of the verb phrase to an A-position below the subject, followed by remnant movement of the verbal predicate above the subject. Note that it is debatable whether the language has a derived A-position for subjects (§4.2); the term 'subject' refers here to the highest DP within the A-domain, typically the external argument.

(4)



[5] The SJPM data are represented in IPA. Tones are represented using the Chao numeric system. For the SJPM data, V$^1$ indicates the lowest tone and V$^5$ is the highest tone (a contextually restricted process of upstep may also produce super-high tones, transcribed as V$^6$). See Caballero et al. to appear for an overview of the tone system of SJPM.

In SSZ, pronominal arguments of a wide range of grammatical functions (e.g., subject, direct or indirect object, object of preposition, possessor) may be expressed as enclitics. As subjects and objects, the clitics also surface in VSO order, both attaching to the verb. Clitics are taken by Sichel and Toosarvandani (2024) to be generated via Agree (following Sportiche 1993, Anagnostopoulou 2003, a.m.o.)—specifically, *pronoun movement* triggered by the Agreeing probe. As also discussed in related earlier work (e.g. Toosarvandani, 2017), SSZ displays the *strong PCC* in monotransitive constructions. Thus, 1st and 2nd (but not 3rd) person clitics in object position are banned, regardless of the higher subject clitic's features. The examples in (5) further show that, to express the PCC-violating configurations, the lower argument must instead be encoded as a tonic pronoun.

(5) a. Bi llre'=**o'** **nada'**
NEG HAB.see=2SG 1SG
'You don't see me.' (*1SG =*a'*)

b. Bi llre'=**la'** **lue'**
NEG HAB.see=1SG 2SG
'I don't see you.' (*2SG =*o'*)

c. Bet=gak=**a'**=**ba'**
CMPL.kill=PL=1SG=3.AN
'I killed them (animals).'
(Yalálag; Foley and Toosarvandani 2022)

SSZ subjects and objects display two additional hierarchy effects. In these constructions, the same 'repair' arises: when cliticization is blocked, the object is encoded as a tonic pronoun. First, combinations of 3rd person clitics display restrictions along a ranking of EL(der)>HU(man)>AN(imate)>IN(animate), in that the former must outrank the latter; this resembles a *strictly descending* (or *ultrastrong*) PCC pattern and is termed the *Gender-Case Constraint* (or GCC) in previous work (Foley et al., 2019; Foley and Toosarvandani, 2022). There is variation within SSZ as to exactly which clitic combinations induce the GCC; only the Yalálag variety is shown in (6).[6] For space reasons, this paper largely abstracts away from the GCC (though see fns. 8, 16, and 17).

(6) a. Bdin=**ba'** **lebe'**
CMPL.bite=3.AN 3.HU
'It (anim.) bit her/him.' (*3.HU =*be'*)
(Foley and Toosarvandani, 2022, p. 12)

b. Wdao=**ba'**=**n**
CMPL.eat=3.AN=3.IN
'It (anim.) ate it (inan.).'
(López and Newberg, 2005, p. 8)

Our focus is on a second hierarchy effect: no object clitic *of any person or animacy* is permitted if the subject is a lexical DP. This was seen in (1) and further illustrated in (7). Sichel and Toosarvandani (2024) frame this as a *Lexical DP Blocking* effect (LDB), the intuition being that the presence of a lexical DP blocks the object from cliticizing. Sichel and Toosarvandani further claim that the LDB is a novel pattern within the existing typology of PCC patterns, though they also observe that it is akin to the *weak PCC*: if a monotransitive contains a single lexical DP argument, it must be in object position.[7]

(7) a. Dz$^{3}$la$^{3}$lle'$^{3}$ **Xwanh$^{1}$=a'$^{3}$** **lhe'$^{1}$**
CONT.forget Juana=DEF 2SG
'Juana forgot you.' (*2SG =*u'$^{3}$*)
(Laxopa; Sichel and Toosarvandani 2024)

b. Blenh$^{3}$ **Xwanh$^{1}$=a'$^{3}$** **leb$^{13}$**
CMPL.carry Juana=DEF 3.AN
'Juana carried it.' (*3.AN =*eb$^{3}$*)

[6] Beyond Yalálag, other SSZ varieties display slightly different versions of the GCC (Foley et al., 2019; Foley and Toosarvandani, 2022; Foley et al., to appear), though all varieties display the LDB. For instance, the Laxopa variety permits HU/EL clitic combinations in either order, though otherwise obeys the HU>AN>IN ranking of Yalálag; the Zoogocho variety is even more permissive: HU/EL and AN/{EL,HU} combinations are licit, though IN/AN is still ruled out.

[7] The weak PCC canonically states that, if there is one 3rd person clitic, it must be in direct object position (cf. Bonet, 1991).

Note that the LDB is not alleviated when the DP subject is displaced (e.g., focus-fronted) to a preverbal position, (8). This reinforces that the LDB is not a purely morphological or prosodic restriction. Rather, the underlying explanation must be structural in nature: in (8), it is the (non-overt) base copy of the DP subject that triggers the LDB.

(8) **Xwanh$^{1}$=a'$^{3}$**$_i$ blenh$^3$ $t_i$ $\boxed{\textbf{leb}^{13}}$
Juana=DEF CMPL.carry 3.AN
'JUANA carried it.' (*3.AN =*eb$^3$*) (Laxopa; Sichel and Toosarvandani 2024)

The PCC and LDB patterns in SSZ may be summed up thus far as (9):

(9) *The PCC/LDB in SSZ (where 1/2/3 are pronominal clitics):*

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Subj. | 2 | 1 | 2 | 1 | 3 | 3 | 3 | DP | DP | DP |
| Obj. | 3 | 3 | 1 | 2 | 1 | 2 | 3 | 3 | 1 | 2 |
| | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✓/✗ | ✗ | ✗ | ✗ |

### *2.2 Subject intervention and a 'Probe Activation' account*

Assuming that pronominal cliticization is preconditioned on Agree, Sichel and Toosarvandani (2024) take well-formed configurations containing two clitic arguments to reflect successful multiple Agree. The Agreeing probe is assumed to be located above both the subject and object but beneath the verb, accounting for the ability for both clitics to move to a position where they can encliticize to the fronted verb. Violations of any of the above hierarchy effects in SSZ result from the probe's failure to Agree with the object, the lower of the two nominals. Given the LDB, this means that a lexical DP subject must also somehow prevent the probe from accessing the object. As the authors observe, the LDB has the general profile of *defective intervention* in A-movement, attested cross-linguistically: in the latter, too, movement of a lower element (Y) triggered by a probe P is blocked by an intervening higher element (X) that itself cannot move, (10).

(10) *P . . . $\boxed{\text{X}}$ . . . Y

Prior work on hierarchy effects appeals to the representations of different features, in that higher ranked features involve more articulated feature geometries than lower ranked ones (Harley and Ritter, 2002; Béjar, 2003, a.o.). To explain why lexical DP subjects participate in such effects, Sichel and Toosarvandani reason that all nominals of category D—pronouns and DPs alike—bear a dedicated D-feature [$\delta$]. All pronouns bear an additional feature [$\pi$] (cf. Sichel and Wiltschko, 2021), and 1st/2nd person pronouns additionally bear [PART] (higher ranked 3rd person pronouns bear additional animacy-denoting specifications of [$\pi$]). The feature geometries for full DPs, 3rd person pronouns, and PART pronouns are presented in (11), with features on the right entailing those on the left.

(11) *Extended Person (Sichel and Toosarvandani, 2024):*
Lexical DPs share a feature ($\delta$) that is part of the structure of Person; only personal pronouns are specified for additional Person features.

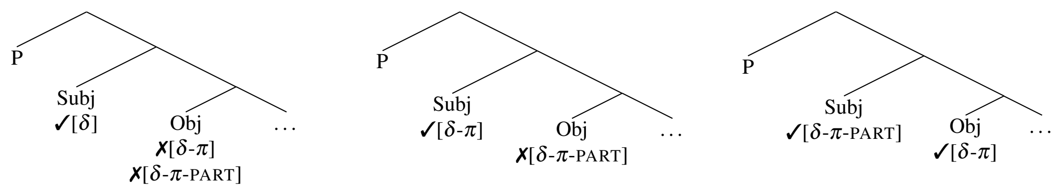
a. *Lexical DP:* [$\delta$]
b. *3rd pers. pronoun:* [$\delta$-$\pi$]
c. *1st/2nd pers. pronoun*: [$\delta$-$\pi$-PART]

Sichel and Toosarvandani propose that the probe, P, is always able to target the subject, the closest goal in P's c-command domain. P may also Agree with the object only if it is *not* more featurally specified than the subject (cf. Anagnostopoulou 2005; Nevins 2007). This is formulated as *Probe Activation*, synthesized in (12). Thus, whereas the initial round of Agree takes place to value P, subsequent rounds of Agree may occur to satisfy requirements of the lower goal G2—so long as G2 does not bear additional features not already found on G1. The creation of any object clitic thus requires Probe Activation.

(12) *Probe Activation (adapted from Sichel and Toosarvandani 2024):* After Agreeing with its highest goal G1, the probe is "activated" and is able to Agree with another goal G2, just in case G2 is not featurally distinct from the probe P: i.e., G1 is not more featurally specified than P for the features P searched for (G2 $\subseteq$ P).

Putting this together with the Extended Person component of their proposal, Sichel and Toosarvandani are able to capture the hierarchy effects in SSZ. As shown in (13a), if the subject is a lexical DP (bearing only [$\delta$]) and the object is a pronoun (of any person), Probe Activation is impossible since the latter bears (at least) [$\delta$-$\pi$]. As such, the object cannot be realized as a clitic. It can, however, be exponed as a tonic pronoun, thus accounting for the 'repair' seen in the examples above. The structure in (13b) illustrates the strong PCC, in that P may generate a 3rd person ([$\delta$-$\pi$]) subject clitic via Agree, but not a participant ([$\delta$-$\pi$-PART]) object clitic. Finally, (13c) illustrates a well-formed two-clitic construction: here, the 3rd person ([$\delta$-$\pi$]) pronoun does not bear additional features not already copied by the subject ([$\delta$-$\pi$-PART]), so an object clitic may be generated by Probe Activation.[8]

(13) a. **DP/pron:* b. **3/*PART*:* c. PART*/3:*




To sum up, Sichel and Toosarvandani (2024) take the LDB to arise from four interacting factors: (i) a high probe P, c-commanding both arguments of the verb, (ii) an Agree requirement on pronominal cliticization, (iii) the extension of pronominal feature geometries to lexical DPs, which are featurally less specified than pronouns, and (iv) a model of Agree in which subsequent rounds are permitted only if no new features would be copied to P.

As mentioned above, Sichel and Toosarvandani offer the LDB pattern in SSZ as a novel extension of the PCC, framed in terms of defective intervention. Below, we present data from the San Juan Piñas variety of Mixtec (SJPM), showing that SJPM also displays the LDB alongside canonical PCC effects in monotransitives. However, the exact patterns that arise in SJPM differ from SSZ in analytically informative ways.

## 3 Extension to San Juan Piñas Mixtec

### *3.1 Language background and key morphosyntactic facts*

Like Zapotec, Mixtec an Eastern Otomanguean language. Though Zapotec and Mixtec are categorized in distinct subgroups within Eastern Otomanguean, they share several morphosyntactic similarities relevant to

[8]This system also extends straightforwardly to the GCC, if animacy features are represented geometrically as additional specifications of [$\pi$], with higher-ranked features entailing lower-ranked ones (e.g., 'elder' may be represented as [$\delta$-$\pi$-AN-HU-EL]; 'inanimate' as [$\delta$-$\pi$]). Probe Activation is possible only if the object does not outrank the subject in animacy. Lastly, to capture ill-formed *PART/PART combinations also characteristic of the strong PCC (e.g., *1/2 and *2/1), Sichel and Toosarvandani appeal to additional features such as [SPKR] that differentiate 1st from 2nd person. To be precise, 1st and 2nd person are specified as [$\delta$-$\pi$-PART-SPKR] and [$\delta$-$\pi$-PART], respectively. 2/1 configurations thus are ruled out the same way as 3/PART, as in (13b), with the conditions for Probe Activation not being met. However, *1/2 configurations require an additional stipulation: the [PART-SPKR] portion of the feature geometry of 1st person is copied back to the probe as a complex unit. This means that [PART-SPKR] (in the representation of 1st person) and [PART] (in the representation of 2nd person) are not only featurally distinct, but the latter is technically *not* a proper subset of the former. As such, Sichel and Toosarvandani take Probe Activation to not be possible in such configurations either.

this paper. SJPM is a Southern Baja variety of Mixtec (Josserand 1983) and is spoken in the town of San Juan Piñas, Santiago Juxtlahuaca, Oaxaca, as well as in diaspora communities in the US. This paper comes out of a broader collaborative project documenting, analyzing, and creating linguistic resources for this previously undescribed Mixtec variety, as overseen by the second author of this paper, an L1 speaker of SJPM. The data in this paper resulted from targeted elicitation with the second author, though the key empirical patterns were also confirmed with three other speakers of SJPM. The paper's analytical and theoretical claims were developed by the first author.

Pronominal arguments in SJPM may surface as enclitics in the same syntactic environments as in SSZ (§2.1). SJPM also displays base V(P)-S-O word order, as in (4) (AUTHORS). And, like in SSZ, SJPM subject/object clitic combinations display the strong PCC, in that 1st and 2nd person (PART) objects must surface as tonic pronouns, not clitics, (14a-b); in contrast, there is no restriction on 3rd person object clitics, (14c). There is no GCC in SJPM, so all combinations of 3rd person clitics are possible.

(14) a. ʃi$^{13}$ni$^{31}$=**$^{n}$do$^{5}$** $\boxed{\mathbf{^{n}du^{1}(^{ʔ}u^{1})}}$
CMPL.see=2PL 1PL.EX
'You (pl.) saw us (ex.).'
(*1PL.EX =*$^{n}$di$^{3}$*)

b. ko$^{5}$ni$^{1}$=**ɲa$^{5}$** $\boxed{\mathbf{^{n}du^{1}(^{ʔ}u^{1})}}$
CONT.love=3SG.F 1PL.EX
'She loves us (excl.).'
(*1PL.EX =*$^{n}$di$^{3}$*)

c. ko$^{5}$ni$^{1}$=**$^{n}$di$^{1}$=na$^{1}$**
CONT.love=1PL.EX=3PL.N
'We (excl.) love them.'

Importantly, this restriction still holds even the subject is a lexical DP, (15).[9] The example in (15b) additionally mirrors (8) from SSZ above; the restriction is not alleviated by Ā-extracting the lexical DP subject. Using the same logic as Sichel and Toosarvandani 2024, the fact that non-overt elements (lower copies of movement) induce in this effect suggests that the restriction is syntactic, rather than prosodic or phonological, in nature.[10]

(15) a. ko$^{5}$ni$^{1}$ [ ti$^{5}$ vi$^{3}$lu$^{5}$ ] $\boxed{\mathbf{^{n}du^{1}(^{ʔ}u^{1})}}$
CONT.love CL.3.AN cat 1PL.EX
'The cat loves us (excl.).' (*1PL.EX =*$^{n}$di$^{3}$*)

b. **ni$^{3}$=ĩĩ$^{3}$=na$^{1}$**$_{i}$ ko$^{15}$-ʃi$^{3}$ni$^{31}$ $t_i$ $\boxed{\mathbf{^{n}du^{1}(^{ʔ}u^{1})}}$
NEG=one=3PL.N NEG.CMPL-see 1PL.EX
'No one saw us (excl.).' (*1PL.EX =*$^{n}$di$^{1}$*)

Thus, we conclude that SJPM, like SSZ, displays the LDB. However, the restriction found in SJPM diverges from that in SSZ in an important respect: it only affects PART *objects*, so combinations of DP subjects and

[9]While this paper focuses on SJPM, the inability for 1st/2nd person objects to be expressed as clitics (regardless of the subject) appears to be a broader property of Mixtec, having been cursorily mentioned in descriptive works on the Chalcatongo (Macaulay, 1996, pp. 140-141), Alacatlatzala (Zylstra, 2012, p. 64), Alcozauca (Cline, 2018, p. 24), Ixtayutla (Penner, 2019, pp. 26, 52-54), San Sebastián del Monte (Mantenuto, 2020, pp. 64-72), and Ocotepec (Alexander 1988, pp. 263-265, Lopez 2022, pp. 54-57) varieties. None of these descriptions explicitly frame such restrictions as the PCC (or LDB).

[10]Related to this, a reviewer points out that, since clitics do not overtly move in SJPM, the argument that cliticization is syntactic rather than phonological in SJPM is weaker than in SSZ. However, there is additional (indirect) evidence internal to SJPM that the pronominal clitics are syntactically generated. The [PART] pronouns in SJPM may be exponed as pronominal clitics, weak pronouns, or strong pronouns (e.g., 1PL.EX *ndi$^{3}$* vs. *ndu$^{1}$* vs. *ndu$^{1}$ $^{ʔ}$u$^{1}$*, as shown in (14)). The weak pronouns resemble the pronominal clitics morphophonologically in that they are also monomoraic (as opposed to bimoraic) and prosodically deficient. However, the weak pronouns pattern like the strong pronouns *morphosyntactically* in that they are not restricted by the PCC or the LDB (i.e., the PCC/LDB 'repair' may involve a weak or a strong pronoun in lieu of a clitic in object position). This, in turn, suggests that the inability for [PART] clitics to be generated in object position cannot be attributed to prosodic or phonological constraints, leaving syntax as the remaining culprit.

3rd person object clitics are permitted, (16). Thus, whereas the LDB in SSZ resembles a weak PCC pattern (§2.1), its counterpart in SJPM looks like the strong PCC. In (16), we also see that, in licit DP/pronoun combinations, the object pronoun simply encliticizes to the immediately preceding element. In verb-initial word order, the clitic host is the postverbal subject, but, if the subject is Ā-extracted, then the object clitic simply attaches to the verb.

(16) a. ko$^5$ni$^1$ [ **ti$^5$** **vi$^3$lu$^5$** ]=**ra$^3$**
CONT.love CL.3.AN cat =3SG.M
'The cat loves him.'

b. **ni$^3$=ĩ$^3$ĩ$^3$=na$^1$** ko$^{15}$-ʃi$^3$ni$^{31}$=**na$^1$**
NEG=one=3.PL NEG.CMPL-see=3.PL
'No one saw them.' ni$^3$=ĩ$^3$ĩ$^3$=na$^1$ ko$^{15}$-ʃi$^3$ni$^{31}$ $t_i$ =na$^1$

The PCC and LDB in SJPM monotransitives are summarized so far in (17), with the divergences from the SSZ pattern $\boxed{\text{boxed}}$ (cf. (9)).

(17) *The strong PCC/LDB in SJPM (where 1, 2, and 3 are clitics):*

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Subj. | 2 | 1 | 2 | 1 | 3 | 3 | 3 | DP | DP | DP |
| Obj. | 3 | 3 | 1 | 2 | 1 | 2 | 3 | 3 | 1 | 2 |
| | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | $\boxed{✓}$ | $\boxed{✓}$ | ✗ | ✗ |

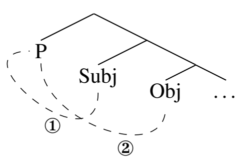
A reviewer asks whether a common analysis of SSZ and SJPM is warranted to begin with: do the SJPM data truly reflect a SSZ-style LDB effect, as opposed to a general restriction on [PART] clitics in object position? In SSZ, the fact that the LDB co-exists with two other distinct hierarchy effects (the strong PCC and GCC) is itself evidence that the LDB is another hierarchy effect, but this evidence is not as clear cut in SJPM. However, there are empirical and conceptual arguments for developing a unified treatment. First, besides the various morphosyntactic similarities already noted, there is another parallel: in both languages, the hierarchy effects are *exceptionally obviated* in double object constructions. This will be detailed in §5, but we mention it here because it is otherwise too big a coincidence to ignore (also, in §5, we specifically show that [PART] direct object clitics are in fact licit in SJPM ditransitives, ruling out a simpler alternative whereby [PART] object clitics are simply disallowed in object position). More broadly, assuming a common syntax for SSZ and SJPM yields new analytical directions *that would be lost otherwise*. Recall Sichel and Toosarvandani's (2024) claim that the LDB is otherwise unattested within the family of PCC effects. As Mixtec and Zapotec are not very closely related, granting that both languages display the LDB widens its potential empirical range to other Eastern Otomanguean languages (see fn. 9 on other Mixtec varieties). Subsequent work on such languages (possibly revealing additional points of variation) may shed further light on the underpinnings of the LDB as a whole. Further, taking the SSZ and SJPM patterns as extensions of the weak and strong PCC, respectively, allows us to more comprehensively evaluate existing theories of the PCC than if SSZ were the sole case study. Finally, the exact SJPM pattern turns out to be identical to Postal's (1989) Fancy Constraint, found in *faire-infinitif* causatives in French and related languages. In such constructions, lexical DP causees may participate in the strong PCC, preventing [PART] but not 3rd person direct object clitics, (18). The SJPM data thus reveal a novel connection between the LDB and the Fancy Constraint, not noticed in Sichel and Toosarvandani 2024. Although space restrictions prevent us from discussing the latter further (though see fn. 22), this also raises the question of whether analyses of the Fancy Constraint (e.g. Sheehan, 2020) can accommodate the LDB.

(18) Marcel **l'** / $\boxed{\textbf{*vous}}$ a fait épouser **au** **médecin**
Marcel 3.ACC= 2PL.ACC= has made marry to.the doctor
'Marcel had the doctor marry her / *you.' (French; Sheehan 2020, pp. 151-152)

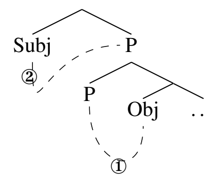
But if we pursue a unified account of the LDB in SSZ and SJPM, we must explain their differing clitic positions, since SJPM object clitics remain *beneath* a higher lexical DP. We suggest that the clitics' surface positions *do not necessarily* indicate the height of the probe Agreeing with them. If clitic position transparently corresponded to probe height (as often assumed in previous work, starting with Kayne 1975), this would suggest a high probe P in SSZ (and French) that asymmetrically c-commands all relevant arguments, (19a), and a lower P in SJPM sandwiched between the subject and object, (19b). By hypothesis, P in the latter case would asymmetrically c-command just the object, but be able to target the subject in P's specifier after cyclic expansion (Béjar and Rezac, 2009).

(19) *Multiple Agree with high P:*

a.




b. *Multiple Agree with low P:*

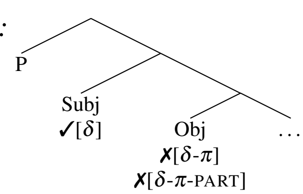



But there are reasons to reject (19a-b) as co-existing derivations in SSZ and SJPM. This direction would force us to develop two distinct analyses of the LDB in SSZ and SJPM, which we wish to avoid for reasons outlined above. Similarly, it denies any analytical unification between SJPM and the French-type Fancy Constraint. Overall, this paper submits that it is more fruitful to maintain a uniform system of Agree in SSZ and SJPM, with differences in hierarchy effects amounting to (micro)parameterization and independent (possibly non-syntactic) factors determining where the clitics ultimately surface. We demonstrate that this is possible for both Sichel and Toosarvandani's account (§3.3), as well as an alternative building on Deal 2024 (§4).

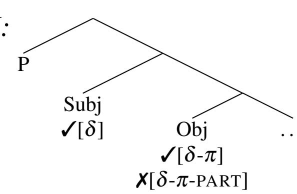
### *3.2 A Probe Activation account of SJPM*

Recall that Sichel and Toosarvandani's (2024) account involves a high probe P c-commanding all arguments, a Probe Activation model of multiple Agree, and the featural representations from the authors' Extended Person hypothesis. To fold the SJPM pattern into this analysis, we may simply assume that P both Agrees and triggers clitic movement in SSZ, but *only Agrees* in SJPM, which is why SJPM clitics remain low. Given these premises, however, the exact LDB pattern seen in SJPM presents an apparent challenge. A lexical DP ([$\delta$]) subject should prevent the probe P from targeting 1st/2nd person ([$\delta$-$\pi$-PART]) *and* 3rd person ([$\delta$-$\pi$]) pronominal object clitics alike, since all pronominal clitics are more featurally specified than lexical DPs. Recall, however, that DP/3 combinations are in fact licit in SJPM. That SJPM bans *DP/PART combinations yet permits DP/3 combinations, as shown in (20b), is thus contradictory under this account.

(20) a. *LDB in SSZ:*

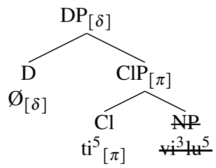



b. *LDB in SJPM:*

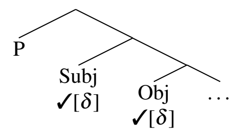



A circumvention of this issue, we suggest, is to treat 3rd person 'clitics' in SJPM as underlyingly *lexical DPs*—an idea that turns out to be independently supported in SJPM. As already seen in examples above, lexical DPs contain prenominal classifiers that are identical to their pronominal counterparts (e.g., *ti*$^5$ *vi*$^3$*lu*$^5$ 'the cat' vs. *ti*$^5$ 'it (anim.)'). In contrast to the [PART] pronouns, which have dedicated clitic and tonic pronoun forms, there are no 3rd person tonic pronouns in the language, so the clitic/classifier form is the only way to expone 3rd person pronominal meaning. Put together, it is not implausible that 3rd person 'pronouns' in SJPM are in fact lexical DPs with elided (or null) NP complements, in the sense of, e.g., Elbourne 2005. This is illustrated in (21a). Under this approach, the differences between SSZ and SJPM amount not to any differences in probe specification, but simply in the underlying representation of 3rd person pronouns. DP/3 configurations are ruled out in SSZ due to Probe Activation, but in SJPM these configurations actually consist of two [$\delta$]-bearing lexical DPs. There is then no LDB to begin with, as in (21b).[11] Lastly, recall that there is no GCC in SJPM. This too follows from the present idea, if animacy features are subfeatures of [$\pi$], not [$\delta$] (see fns. 8 and 17).

(21) a. *SJPM 3rd person clitic as DP:*




b. *Reanalysis of SJPM DP/3 based on (21a):*




A potential drawback to this solution is that it may not generalize beyond SJPM. It predicts that, for *any* language that displays the SJPM-type LDB pattern, i.e., when the lower argument is [PART] but not 3rd person, we should be able to independently motivate an analysis of (apparent) 3rd person clitics along the lines of (21b). But it is not clear whether this is borne out. For instance, San Sebastián del Monte Mixtec (SSMM), which is very closely related to SJPM, has an identical PCC and LDB profile to SJPM, (22), but this variety's classifiers are distributionally restricted and co-occur with only a subset of nouns (Mantenuto, 2020). It remains to be seen whether 3rd person clitics in languages like SSMM can be analyzed as in (21a); if not, then the well-formedness of the DP/3 combination in (22b) remains problematic for Sichel and Toosarvandani 2024. Similar considerations are needed for the Romance languages that display the Fancy Constraint of (18).

(22) a. Kàni Juan **yo'o**
CMPL.hit Juan 2SG
'Juan hit you.' (*2SG =*ó*)

b. Sàsi Juan=**ti**
CMPL.eat Juan=3.AN
'Juan ate it.'
(SSMM; Mantenuto 2020, pp. 68, 70)

In sum, SJPM displays a strong PCC patterning in monotransitives, regardless of whether the subject is a clitic or a lexical DP. That DP/3 combinations are well-formed in SJPM (as well as in other languages mentioned above) is unexpected under the analysis of Sichel and Toosarvandani 2024, which wholly rules such combinations out. A possible workaround was suggested, whereby 3rd person clitics in SJPM (but not SSZ) are underlying lexical DPs, though the cross-linguistic validity of this idea is still to be determined. Regardless, we take the existence of the SJPM-type patterning to be an important wedge for assessing Sichel and Toosarvandani's analysis, as well as for competing analyses.

## 4 Evaluating an alternative: Deal 2024

We now offer an alternative account of the LDB, building on the theory of PCC effects developed by Deal 2024. We focus on this particular implementation because it too takes the PCC to result from the inability for a single probe P to Agree with both arguments, and because it is explicitly named by Sichel and Toosarvandani 2024, pp. 34-36 as unable to simultaneously account for the PCC, GCC, and LDB. This is because, unlike pronominal clitics, lexical DPs are not generated by Agree to begin with, so failure of P to

[11] At this point, an alternative explanation for the low position of SJPM object clitics comes to mind: if 3rd person pronouns are lexical DPs, then their non-movement follows from the simple fact that lexical DPs do not move either. However, while this captures the monotransitive data shown so far (as low object clitics are only visible in the 3rd person), this cannot account for the ditransitive data to be discussed in §5, as we will see that [PART] clitics also remain in situ.

Agree with a lexical DP subject should not necessarily result in ungrammaticality.[12] We show that an Interaction/Satisfaction analysis can in fact accommodate the core LDB patterns, and can do so without adopting any additional assumptions about the status of 3rd person clitics. However, this analysis does not maintain any logic of defective intervention, and requires non-trivial assumptions about Eastern Otomanguean clause structure more generally.

### *4.1 An Interaction/Satisfaction account of SJPM and SSZ*

In the theory of Agree developed in Deal 2015, probes are specified for *interaction* and *satisfaction* conditions (henceforth, 'Int/Sat'), dictating which features are copied back to a probe P, vs. which ones halt further probing by P. Deal 2024 extends this to the PCC: taking pronominal clitics to require Agree, the PCC arises when P encounters a goal with features that match P's satisfaction condition, preventing P from interacting with a second potential goal. Different PCC effects thus arise from different Int/Sat specifications of P.

Importantly, this account requires that P interact with the direct object first and the higher argument (e.g., the subject) after that. This can be achieved if the head containing P also introduces the subject; after P Agrees with the object in its c-command domain, its search domain cyclically expands to include its specifier (Béjar and Rezac, 2009). This is the configuration depicted in (19b), already empirically motivated in SJPM given the low position of object clitics. However, under this account, the LDB can no longer be understood in terms of subject intervention (in other words, the term 'Lexical DP Blocking' becomes a misnomer). This is because the probe P may never encounter the subject before the object; rather, the object's features block subsequent Agree with the subject. Losing the parallel with defective intervention effects cross-linguistically is a potentially undesirable consequence of this reframing, though in §4.2 we offer a novel theoretical connection in a different morphosyntactic area.[13]

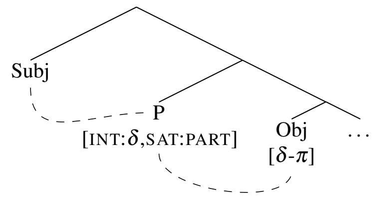
For Deal, the strong PCC arises when the probe P bears [INT:$\pi$,SAT:PART]. If the object is 3rd person ([$\pi$]), then P simply copies this feature and continues probing, but if the object bears [PART], P is satisfied and stops probing further. Although Deal's theory is designed to account for combinations of pronouns, we show that it can also handle the LDB. Since lexical DPs bear [$\delta$] but not [$\pi$] per Sichel and Toosarvandani 2024, let us assume that, for any language that displays the LDB, P bears [INT:$\delta$,SAT:PART] rather than [INT:$\pi$,SAT:PART].[14] The trees in (23) illustrate how this system applies to the strong PCC-like LDB pattern of SJPM. If the object is a 3rd person pronoun ([$\delta$-$\pi$]) or a lexical DP ([$\delta$]), P may continue to probe for its specifier, the subject, (23a); however, if the object bears [PART], P is satisfied and does not probe for the subject, (23b) (why non-agreement with the subject ultimately crashes the derivation, as indicated with '☹', is addressed in §4.2). In contrast to Sichel and Toosarvandani's (2024) analysis, this alternative does not require any assumptions about the underlying syntax of 3rd person pronouns in SJPM: they may be represented as genuine pronouns ([$\pi$]) or as Ds with elided NP complements ([$\delta$]).

[12] For a similar reason, Sichel and Toosarvandani 2024 argue that Coon and Keine's (2021) Feature Gluttony approach to the PCC cannot capture the LDB either. Due to space constraints, this paper cannot discuss this account further, but the considerations discussed in §4.2 below may apply there as well.

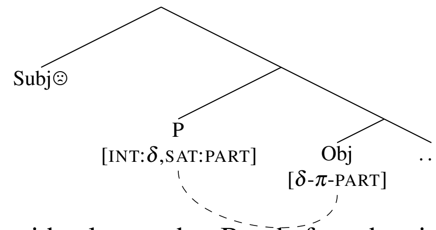
[13] A reviewer asks whether we may maintain a high P under an Int/Sat analysis, or a low P within Probe Activation (assuming that probe height and mechanism of Agree are independent properties of these accounts and may in principle be crossed in a 2x2 typology). However, both of these options predict *reverse* hierarchy effects (whereby the lower argument outranks the higher argument), unattested in both SSZ and SJPM. See Arregi and Hanink 2022 and Deal 2024, pp. 77-79 on such reverse patternings within an Int/Sat framework.

[14] Empirically, the LDB (in both languages) can also be derived by maintaining an interaction condition as in [INT:$\pi$], so that lexical DPs are never targeted by P. This is closer to Sichel and Toosarvandani's (2024) position that lexical DPs do not Agree. The choice to specify the interaction condition as [int:$\delta$] is primarily in service of the extension of the analysis in §4.2, wherein it is crucial that lexical DPs *are* accessible to P.

(23) a.




b.




The SSZ pattern may also be derived using an Int/Sat model with a low probe. But, before showing this, we briefly revisit the apparently high locus of pronominal cliticization in the language, as posited by Sichel and Toosarvandani. That object clitics move above lexical DP subjects in SSZ has not been shown, since all such examples have been ungrammatical (in SSZ, no object clitic may co-occur with a lexical DP subject). Overt evidence for this high position instead comes from other contexts. First, focused [PART] (though not 3rd person) subject pronouns are obligatorily doubled. In monotransitives consisting of a focused clitic-doubled subject and a pronominal object, the object indeed raises past the *in situ* subject to attach to the verb, (24a). Second, as noted in §3.1, the LDB is (partially) obviated in ditransitives in SSZ (§5); assuming base IO>DO order, (24b) shows that a direct object clitic may move over a lexical DP (the indirect object).[15]

(24) a. Betw=a'=**ba'**$_i$ **neda'** $t_i$
CMPL.hit=1SG=3.HU 1SG
'I hit her/him.' (Sichel and Toosarvandani, 2020)

b. Blu'i=b=**ne'**$_i$ **me'edo'** $t_i$
CMPL.give=3.AN=3.EL baby
'It (anim.) showed the baby her/him (eld.).' (Laxopa; Foley and Toosarvandani 2019, p. 256)

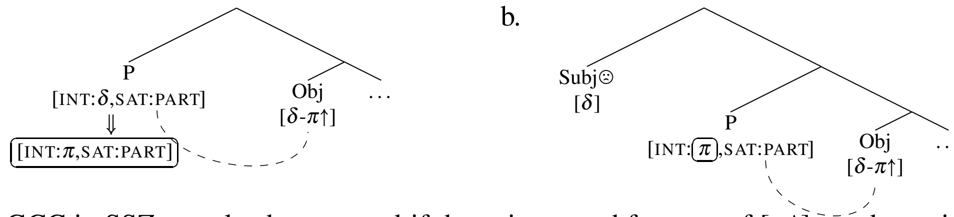
That being said, it may be possible to analyze examples like (24) without *syntactic movement*. For instance, as clitic displacement is known to often be governed by morphophonological factors (e.g. Zribi-Hertz and Diagne, 2002; Bennett et al., 2016), it may be that the clitics in SSZ are simply subcategorized to attach to the verb stem rather than a preceding DP (in contrast, SJPM would have no such requirement). An assumption along these lines would be needed to maintain that clitics in both SSZ and SJPM are generated by a uniformly low probe, and we may expect this to be corroborated or refuted by independent prosodic or morphophonological properties of the language (cf. Brinkerhoff et al. 2021).

Turning now to an Int/Sat analysis of SSZ, recall that, while the subject and object clitics display the strong PCC (as well as the GCC), the LDB in SSZ resembles the weak PCC. In Deal 2024, the weak PCC is captured using the notion of *dynamic interaction*: a target of Agree may change the interaction conditions of the probe in the course of the derivation. Deal proposes that languages with the weak PCC use the same condition [INT:$\pi$,SAT:PART] as for the strong PCC, but the [PART] feature is dynamic ([PART$\uparrow$]): if P encounters a goal bearing [PART$\uparrow$], then it gets updated to [INT:PART,SAT:PART]. Since SSZ also displays the strong PCC in clitic combinations, we may assume the same probe specification as for SJPM above: [INT:$\delta$,SAT:PART]. In SSZ, however, the feature [$\pi$] would be dynamic (annotated as [$\pi\uparrow$]). If the object bears [PART], then P is satisfied in the first cycle of Agree (the pattern in (24b)). But if the object bears [$\pi\uparrow$], then P gets updated to [INT:$\pi$,SAT:PART], (25a). This does not impact Agree with pronominal subjects ([$\pi$])—but does prevent Agree with lexical DP ([$\delta$]) subjects, (25b), thus deriving the LDB.

[15] Foley and Toosarvandani (2022) also offer an indirect argument for clitic movement, based on the observation that pronominal clitics cannot be formed in coordinate structures. The authors analyze this pattern as a CSC island violation, thus implicating syntactic movement. However, we suggest this restriction could instead be due to prosodic or morphophonological requirements on coordinands, since we independently know from work such as Cardinaletti and Starke 1999 that prosodically deficient pronominal elements generally cannot appear in coordinate structures regardless of whether they move. For what it is worth, in SJPM, no pronominal clitics or weak pronouns (cf. fn. 10) of any person may be coordinated.

(25) *LDB in SSZ:*

a. b.




Finally, the GCC in SSZ can also be captured if the animacy subfeatures of [$\pi\uparrow$] are dynamic as well (i.e., [EL$\uparrow$], [HU$\uparrow$], [AN$\uparrow$]). This rules out any 3/3 combination in which the object outranks the subject (see also Deal and Royer 2025). To explain its absence in SJPM, we only have to add that these features are not dynamic in SJPM.[16]

At this point, a question arises: if Agree is needed for cliticization (as Deal also assumes), Agree with only the object should prevent generation of a clitic in *subject position*. Why, then, is it only the object pronoun that surfaces as a tonic pronoun in both languages? An illustration of this asymmetry is given in (26) with SJPM.

(26) a. *ko$^5$ni$^1$ $\boxed{\textbf{ⁿdo}^{5}\textbf{(ˀo)}^{1}}$=**ⁿdi**$^{\mathbf{1}}$
CONT.love 2PL=1PL.EX
Intended: 'You (pl.) love us (excl.).'

b. ko$^5$ni$^1$=**ⁿdo**$^{\mathbf{5}}$ $\boxed{\textbf{ⁿdu}^{1}\textbf{(ˀu}^{1}\textbf{)}}$
CONT.love=2PL 1PL.EX
'You (pl.) love us (excl.).'

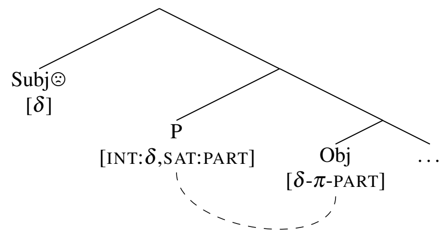
Deal 2024 leaves the issue of PCC 'repairs' open, though one possibility offered therein is that pronominal clitics and tonic pronouns are not derivationally related (p. 56); either could in principle be Merged in a given derivation, modulo certain grammatical pressures to insert one over the other. Here, we develop a version of this idea. Suppose that pronominal clitics are structurally smaller than their tonic pronoun counterparts, and that the grammar prefers inserting a smaller structure for economy reasons, if possible (e.g. Cardinaletti and Starke, 1999). Specifically, suppose that tonic pronouns contain a DP shell, making them featurally akin to lexical DPs ([$\delta$]). This is independently motivated: in SSZ, Merging a tonic pronoun in subject position (required when the subject is focused) produces a LDB violation, even when the subject's features should outrank the object's in animacy (i.e., there is no GCC violation), (27).[17]

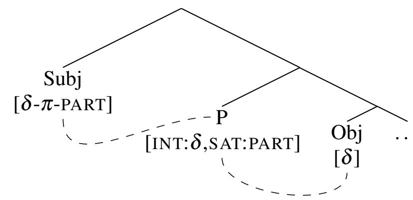
[16]Recall that the GCC resembles the strictly dynamic PCC, except with animacy features at play. In Deal 2024, the strictly descending PCC is derived with the probe specification [INT:$\pi$,SAT:SPKR], coupled with the dynamic nature of the [PART$\uparrow$], as also discussed above for the weak PCC. This means that, if the direct object is [SPKR] (which entails [PART]), P cannot also Agree with the subject at all; if the direct object is 2nd person ([PART] without [SPKR]), then P can only Agree with the subject if it is 1st person (assuming that 2/2 combinations are independently ruled out). Since the GCC only affects combinations of 3rd person pronouns (there is no ranking between 1st and 2nd person), we may continue to use our proposed [INT:$\delta$,SAT:PART] probe condition, coupled with the dynamicity of [$\pi$] and its animacy subfeatures. Consider, for instance, an illicit AN/HU combination. If P interacts with a dynamic [HU$\uparrow$] feature, then P is updated to [INT:HU,SAT:PART]. P cannot then Agree with an animate subject, which lacks [HU]. In fn. 6, it was also mentioned that the exact manifestation of the GCC differs across SSZ varieties. Under the present Int/Sat analysis, we can account for this variation if, across SSZ, [$\pi\uparrow$] is invariably dynamic (accounting for the LDB), but there is variation in exactly which animacy subfeatures of [$\pi$] are also dynamic.

[17]A reviewer correctly points out that many languages do permit Agreement with tonic pronouns; indeed, Deal 2024 shows that, in some languages, tonic pronouns also may participate in the PCC. A potentially relevant connection is that, whereas SSZ lexical DPs do not display animacy restrictions (despite encoding nominals of varying animacy specifications), they do in other languages. For instance, Deal and Royer 2025 show that, in contrast to SSZ, combinations of lexical DPs are ruled out in certain Mayan languages whenever the object outranks the subject. One possible direction might be to parameterize whether the relevant person/animacy features are *properly contained* within the DP, such that they are not visible to any DP-external probes, or are located in the outermost nominal layer. This seems related to prior work on morphosyntactic variation in terms of splitting features across different heads vs. bundling them within a single head (e.g. Giorgi and Pianesi, 1997; Scontras et al., 2018; Hsu, 2021).

(27) *Blhe'e=**b**$_i$ **leba'** $t_i$
CMPL.see=3.AN 3.HU
Intended: '<u>S/he</u> saw it (anim.).' (Sichel and Toosarvandani, 2020)

Thus, in order to encode any subject/object combination that violates a hierarchy restriction (e.g., 3/PART), a tonic pronoun is Merged in lieu of a clitic, overriding the aforementioned economy-driven preferences. But the tonic pronoun *must* be Merged in object position, given the existence of the LDB, as (27) attests. As illustrated in (28a), based on (26a), inserting the ([$\delta$]) tonic pronoun in subject position would yield a LDB violation in the same way as a lexical DP subject would. In contrast, Merging it in object position feeds second-cycle Agree, allowing P to also Agree with its specifier, the subject, (28b).[18]

(28) a.



b.



So far, we have shown that the Int/Sat model of Agree (Deal, 2024) can generate the core hierarchy patterns in SJPM and SSZ, if both languages have a low probe P specified for [INT:$\delta$,SAT:PART] that encounters the object before (possibly) Agreeing with the subject. Under this approach, the crucial point of variation is that SSZ has dynamic [$\pi$] features that may change P's interaction specification in the course of the derivation. But not yet addressed is why the inability to Agree with a lexical DP (or tonic pronoun) subject should yield ungrammaticality to begin with. We address this next.

### *4.2 Implications for nominal licensing and clause structure*

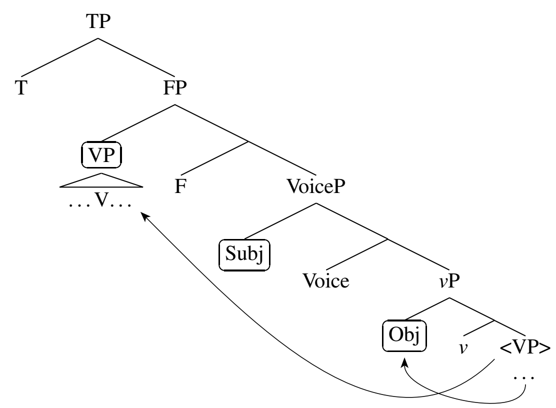
As pointed out in Sichel and Toosarvandani 2024, the LDB does not automatically fall out from Deal 2024's system: lexical DPs are not generated by Agree, so, unlike clitics, they do not *need* to be Agreed with by the low probe P responsible for hierarchy effects. We concur with this logic, but we believe this issue to be surmountable. For Deal's (2024) approach to work, we must simply identify some *additional* property of SSZ and SJPM that enforces this requirement. We suggest that the relevant property may be *verb-initial word order* (as first mentioned in §2.1), derived by object shift and remnant VP-fronting to a pre-subject position. If correct, this offers independent motivation for developing an Int/Sat analysis of the LDB. As alluded to above, such an analysis abandons Sichel and Toosarvandani's view that the LDB involves defective intervention (indeed, Deal's theory generally departs from this view of PCC effects, contra Anagnostopoulou 2003, 2005 and others). As a result, we potentially lose this deeper explanation for why the LDB arises to begin with. But if the LDB may be understood as directly stemming from general properties of SSZ and SJPM clause structure, then this helps to relevel the playing field between these accounts.

If we assume that the object raises to Spec-*v*P and the subject is generated in a higher Spec-VoiceP (Pylkkänen, 2002; Harley, 2013), we may update the schema in (4) as (29). Thus, continuing with Deal's (2024) approach, our low probe P responsible for the PCC is in Voice.[19] We suggest that all nominal

[18]A complication for this account comes from examples like (24a), which show that focused [PART] subjects are necessarily clitic-doubled in SSZ. In contrast to (27), the doubled [PART] tonic pronoun subject does not behave as a lexical DP for LDB purposes, since the 3rd person object is able to cliticize. One possibility is to take this construction to involve something like *Clitic Right Dislocation*, with the focused tonic pronoun occupying a right-peripheral clausal position. While this is easily disproveable, it is not contradicted by any existing examples in Sichel and Toosarvandani 2024 and related works, which all feature the doubled pronoun is in sentence-final position.

[19]Let us also assume that the head F, which triggers VP-movement to its specifier in (29), is a distinct head beneath T, which hosts tense (also proposed by van Urk 2024 for other verb-initial languages): in both SSZ and SJPM, tense is realized at the left

licensing heads may be located *within the VoiceP domain*. If so, then the LDB is not just due to P's inability to Agree with a lexical DP subject, but also the inability for a subject to interact with *any* Agreeing head in the course of the derivation besides P. Under this view, the LDB results from an interaction between the PCC and the Case Filter, the abstract requirement that all DPs are case-licensed by some A-probe via Agree (Chomsky 2000, cf. Vergnaud 1977).

(29)



Much prior work on verb-initial languages has sought to parameterize the probe specification of T (or F in (29)) so that it seeks a verbal element rather than a DP, one consequence being that DPs may only be Agreed with by VoiceP-internal heads (Alexiadou and Anagnostopoulou, 1998; Oda, 2005; Potsdam, 2009; Doner, 2019). Here, we utilize this idea to SJPM. We focus just on SJPM as a proof of concept, but it may be possible to extend this to SSZ as well (see below). Let us (maximally generously) assume that a possible DP-licensor could be any functional head bearing an A-probe, e.g., bearing case or $\phi$-features, or triggering A-movement. Since SJPM does not display morphological case on DPs or $\phi$-agreement morphology (beyond the pronominal clitics), we must diagnose the occurrence of licensing with A-movement and, of course, the PCC and LDB.

Interestingly, SJPM lacks any *positive* evidence for A-probes outside of VoiceP. It seems to wholly lack expletive subjects and constructions that could be analyzed as raising-to-subject;[20] as exemplified in (30), SJPM speakers routinely provide paraphrases in direct translation tasks. Additionally, while there are unaccusative subjects in SJPM, these remain obligatorily postverbal (again, see (30)), so they also provide no evidence for VoiceP-external licensing heads. Finally, it is not possible to raise a DP to a position between overt C and the VP in Spec-FP, (31), further reinforcing that T lacks an EPP feature.

(30) tu$^{5}$va$^{1ʔ}$a$^{3}$ɾa$^{3}$ ni$^{1}$-kõ$^{3}$õ$^{3}$ sa$^{1}$vi$^{5}$
probably CMPL-fall rain
'Probably rain fell.' (Targeted: 'Parece que llovió' or 'It seems to have rained.')

(31) *a$^{5}$ [ **ɾa$^{1}$** **$^{n}$dʒaʃ$^{51}$**$_{i}$ ] ʃa$^{13ʔ}$a$^{3}$ $t_i$ ĩĩ$^{3}$ ʒu$^{5}$ʃi$^{1}$ni$^{3}$ ka$^{5ʔ}$no$^{3}$ $^{n}$da$^{5}$ ɲa$^{5}$ Clau$^{5}$dia$^{6}$?
Q CL.3SG.M Josh CMPL.give one hat big to CL.3SG.F Claudia
Intended: 'Did Josh give a big hat to Claudia?'
[Second author]'s comment about judgment: 'Ah, no. . . primero "a$^{5}$ ʃa$^{13ʔ}$a$^{3}$."'

edge of the raised verb (whether as grammatical tone or as a segmental morph). Finally, although the basic logic of this section is also compatible with a head movement derivation, as proposed for other Mesoamerican languages (Macaulay, 2005; Clemens and Coon, 2018), in SJPM and closely related varieties there is much evidence that the fronted verbal element is phrasal (e.g. Hedding and Yuan, 2025).

[20] Moreover, as shown in Ostrove 2021 on the closely related San Martín Peras variety of Mixtec, passive constructions are impersonals, so they do not involve A-movement of the theme to a Voice-external position.

Moreover, although short (often string-vacuous) A-movement of subject DPs to a position below the raised verb has been posited for other verb-initial languages (e.g. Longenbaugh and Polinsky, 2018), there is, again, no positive evidence for this in SJPM: for instance, there are no adverbs or overt heads that may intervene between the subject and object (see below). And although DPs may be preverbal in SJPM (in the absence of overt C), these involve Ā-movement, generally associated with quantifier raising or information structure (Macaulay, 2005). Finally, in SJPM we do not find familiar hallmarks of DP-licensing by finite T. SJPM lacks dedicated non-finite morphology, but the so-called 'irrealis' verbal form is used in obligatory control and other contexts that are naturally translated as non-finite (Ostrove, to appear). In such contexts, the embedded subject is necessarily overt.[21] More generally, there are (to our knowledge) no constructions in the language that restrict the distribution of subjects based on TAM distinctions.

(32) a. ku$^5$u$^3$ ɲa$^5$ si$^{5ʔ}$i$^6$ [ ka$^3$ta$^3$**=ɲa$^5$** ]
CONT.try CL.3SG.F woman IRR.sing=3SG.F
'The woman is trying to sing.'

b. ko$^{13}$ni$^1$=ɲa$^5$ [ ka$^3$ʃa$^{5ʔ}$a$^3$ **ɾa$^1$** **X$^w$ã$^5$ã$^1$** ]
CMPL.want=3SG.F IRR.eat CL.3SG.M Juan
'She wanted Juan to eat.'

In contrast, there are A-probes *within* VoiceP. Object shift involves A-movement triggered by *v*; moreover, objects in Spec-*v*P may additionally be Agreed with by the $\phi$-probe in Voice responsible for hierarchy effects. However, a subject Merged in Spec-VoiceP may *only* be targeted by Voice, since it is not in *v*'s c-command domain. But because of the probe's Int/Sat conditions ([INT:$\delta$,SAT:PART]), Agree with the subject may be blocked if P is satisfied by the object's $\phi$-features. Therefore, *any* subject that fails to be Agreed with by Voice violates the Case Filter. Lexical DPs are especially vulnerable to this, given that they are featurally the less specified out of all of the nominal types considered. This, we suggest, would be the source of the LDB under Deal's (2024) account of the PCC.

If a uniform analysis of SSZ and SJPM is to be maintained, there should also be no A-probes outside of VoiceP in SSZ either. We think that this is plausible. As noted in §2, verb-initial word order in SSZ is also derived via object shift and predicate fronting; moreover, existing work on Zapotec have independently generalized that these languages lack at least expletive subjects and passivization, as well as correlations between overt subjects and finiteness (e.g. Sonnenschein, 2004; Lee, 2006; Foreman, 2006). Lastly, in their analysis of SSZ clause structure, Adler et al. (2018) (whom Sichel and Toosarvandani 2024 follow) do posit a short A-movement step of subjects to a position below the fronted VP, motivated by the fact that transitive and unaccusative subjects both precede manner adverbs adjoined to *v*P, (33). However, we believe that the data provided are also compatible with the present account, which diverges from Adler et al. 2018 by decomposing '*v*P' into distinct projections: external arguments are base-generated in VoiceP, objects and unaccusative subjects move to Spec-*v*P, and manner adverbs adjoin to some intermediate verbal projection between *v*P and above the VP constituent that undergoes remnant movement. Thus, the subject>adverb word order seen in (33) need not reflect short A-movement past the adverb at all. For empirical completeness, we add that, in SJPM, *all* manner adverbs front with the VP above external and internal argument subjects, (34). We may account for this difference under the current analysis, if manner adverbs adjoin slightly higher in SSZ (such that they evade VP-fronting) and lower in SJPM.[22]

[21] Ostrove (to appear) analyzes the embedded pronominal subject in (ia) as overt PRO (specifically, pronominal copy control) in closely related San Martín Peras Mixtec.

[22] It may even be possible to analyze the Fancy Constraint in languages like French (§3.1) along these lines. Following Folli and Harley 2007, causees of *faire*-infinitifs are introduced in in Spec-*v*P, directly embedded under the causative verb *faire* (which introduces the causer). Suppose that *v* bears [INT:$\delta$,SAT:PART] just as in SJPM and SSZ, meaning that a causee in Spec-*v*P cannot be Agreed with if *v* is satisfied in its first cycle of Agree. Higher licensing heads such as T do exist in French, but if T only targets the causer (the subject), then the causee would remain licensed—again, violating the Case Filter.

(33) a. U$^3$do$^3$ ***{xti$^1$do'$^1$-yes$^1$}** Jwa$^1$nh=a'$^3$ **{xti$^1$do'$^1$-yes$^1$}** yet$^3$=e'nh$^3$ **{xti$^1$do'$^1$-yes$^1$}**
CMPL.eat quickly-INT Juan=DEF quickly-INT tortilla=DEF quickly-INT
'Juan ate tortillas very quickly.' (Laxopa; Adler et al. 2018, p. 34, Sichel and Toosarvandani 2024, p. 6)

b. Dz-i$^3$-yag$^3$ ***{xti$^1$dao'$^1$}** Pe$^1$dro=nh$^3$ **{xti$^1$dao'$^1$}**
CONT-INCH-be.cold quickly Pedro=DEF quickly
'Pedro is getting cold quickly.' (Guiloxi/Yalina; Sichel and Toosarvandani 2024, p. 6)

(34) a. tu$^5$tu$^5$ ʒu$^{3ʔ}$u$^5$ **{tʃe$^{5ʔ}$e$^5$}** ɲa$^5$ Pa$^3$lo$^3$ma$^5$ ***{tʃe$^{5ʔ}$e$^5$}**
CONT.whistle mouth loudly CL.3SG.F Paloma loudly
'Paloma is whistling loudly.'

b. ka$^1$ʒu$^1$ **{ʒa$^1$a$^1$}** ɲa$^1$ i$^3$ti$^5$ ***{ʒa$^1$a$^1$}**
IRR.burn slowly CL.3.N candle slowly
'The candle will burn slowly.' (SJPM)

Zooming out, this line of analysis posits that the LDB should arise only when highly specific conditions happen to align in a given language: (i) the Case Filter, (ii) a low probe P with a [SAT:PART] condition, and (iii) no other A-probes available to license a DP that P cannot interact with. Its conspiratorial nature may explain the relative rarity of the LDB cross-linguistically. Of course, since condition (iii) is predicated on the total absence of any licensors for the relevant nominal (i.e., negative evidence), finding just one such licensing head would immediately nullify this aspect of the theory. This is not necessarily a negative; rather, it is a clear prediction that can be straightforwardly falsified in future work.

### *4.3 Interim conclusion*

We have shown that the LDB in SJPM can be analyzed uniformly with that in SSZ, with the differences between these languages understood as (micro-)parameterization. Although we could have modeled these points of variation by appealing to more significant syntactic differences, we have argued that maintaining a (near-)uniform syntax ultimately offers a clearer analytical blueprint for future work.

We have systematically applied the accounts of Sichel and Toosarvandani 2024 and Deal 2024 to both languages, and have shown that both accounts are broadly compatible with the data presented thus far. Both accounts utilize Sichel and Toosarvandani's hypothesis that lexical DPs and pronouns share a common feature, [$\delta$]. Sichel and Toosarvandani's analysis derives the LDB from a high probe P that c-commands both arguments, with the possibility of multiple Agree regulated by a mechanism of Probe Activation. In contrast, we have shown that the LDB also follows from Deal's Int/Sat theory of Agree, if coupled with a low P that undergoes cyclic expansion and a licensing requirement that rules out DP subjects that are not Agreed with in the course of the derivation.

The exact differences between SSZ and SJPM also raise some outstanding questions for the two accounts. The LDB in SJPM presents an apparent contradiction for Sichel and Toosarvandani 2024, given that DP/3 configurations are licit; to resolve this, we suggested possibly analyzing 3rd person clitics in SJPM as actually lexical DPs, though it is unclear if this idea can be extended to other languages with identical LDB patterns. For Deal's (2024) theory to uniformly capture SSZ and SJPM, we must assume that the surface position of clitics in SSZ (which invariably encliticize to the fronted verb) is determined postsyntactically, and restrict all possible licensing heads in these languages to the VoiceP-internal domain. As noted above, however, we do not regard these as knock-down problems for either account, but rather testable predictions that may ultimately move the needle closer towards the correct treatment of the LDB, as well as the PCC more generally.

## 5 Extension: Obviation of hierarchy effects in ditransitives

In the remainder of this paper, we turn to ditransitives in SSZ and SJPM, which seem to challenge both analyses. Again, we offer possible analytical directions where relevant, but our goal is to highlight these issues for future research. In earlier work, Foley and Toosarvandani (2019) show that the LDB is exceptionally *obviated* in SSZ ditransitive constructions, in that indirect (IO) and direct object (DO) pairs need not obey this restriction. These facts are briefly mentioned (though not analyzed) in later work by these authors (Foley and Toosarvandani, 2022; Sichel and Toosarvandani, 2024). We show that, once again, SJPM offers a replication of this otherwise surprising pattern, albeit with a twist.

Focusing first on SSZ, (35a) (repeated from (24b)) demonstrates that a DO may cliticize past a higher lexical DP IO, in apparent violation of the LDB. The GCC is similarly obviated in IO/DO clitic combinations, (35b). Importantly, however, the strong PCC still holds, (35c): a [PART] DO may not cliticize regardless of the properties of the IO.

(35) a. Blu'i=b=**ne'**$_i$ **me'edo'** $t_i$
CMPL.give=3.AN=3.HU baby
'It (anim.) showed the baby her/him (eld.).' (Laxopa)

b. Ba blu'i=ba'=**b**=**ne'**
already CMPL.show=3.HU=3.AN=3.EL
'S/he already showed him/her (eld.) it (anim.).' (Laxopa)

c. Elu'ed=ba=**b** $\boxed{\textbf{le'}}$
FUT.show=3.HU=3.AN 2SG
'S/he will show you to it (anim.).' (*2SG =*o'*) (Guiloxi)
(Foley and Toosarvandani 2019, pp. 256, 260, 261)

Foley and Toosarvandani 2019 appeal to the Principle of Minimal Compliance of Richards 1998, which allows a constraint to be lifted so long as it has been obeyed once: the Probe Activation[23] requirement on the high probe P is minimally satisfied after considering just the higher two arguments, the subject and the IO. As such, the DO may cliticize even if it bears additional features not available to the higher arguments. Note, however, that this only accounts for (35a-b); to capture (35c), the authors allude to the Person Licensing Condition of Béjar and Rezac 2003 as a separate constraint that must still be adhered to (Foley and Toosarvandani, 2019, fn. 6). Finally, the authors point out that, for this account to work, there cannot be any additional probes—a single probe must c-command and be able to target all three arguments—and we cannot assume that the expected IO>DO hierarchical order in ditransitives freely alternates with a DO>IO order (that is to say, in an example like (35a), the fact that the DO clitic precedes the IO lexical DP cannot be due to the DO c-commanding the IO to begin with in their base positions).[24]

In this context, SJPM is once again illuminating. First, note that most ditransitives in SJPM are prepositional datives (with IOs enclosed in PPs), so IO/DO clitic combinations do not exist in such constructions. However, there is an instrument applicative construction, (36), in which neither internal argument is prepositional. In this construction, the applied instrument argument necessarily precedes the direct object, regardless of whether it is a lexical DP or a pronominal clitic, (36). Thus, unlike the above SSZ examples, this construction rules out a hypothetical DO>IO structure.[25]

[23] An earlier version of Probe Activation is formulated as the Condition on Pronominal Cliticization in Foley and Toosarvandani 2022.

[24] Focusing just on the pattern in (35a), Sichel and Toosarvandani 2024, p. 50 raise two additional analytical possibilities: perhaps the lexical DP IO is ignored by the probe P, either because IOs are totally invisible (e.g., if they are PPs, not DPs) or because DPs do not require Agree in their system and so they are outside of the purview of P's Probe Activation mechanism. However, these options cannot be unified with the clitic combinations in (35b), not discussed in their paper, since IO clitics do Agree. Thus, we do not consider these ideas further.

[25] However, due to the nature of this construction, the IO (an instrument) cannot be 1st/2nd person. This makes it impossible to

(36) a. kõ$^{13}$õ$^{3}$=na$^{1}$ [ **ɲa$^{1}$ ʒu$^{1}$u$^{5}$ ʒo$^{5ʔ}$o$^{3}$** ]=**ɲa$^{5}$**
CMPL.hit=3PL.N CL.3.N rock this =3SG.F
'They hit her with this rock.'

b. kõ$^{13}$õ$^{3}$=$^{n}$di$^{1}$=**ɲa$^{1}$=ɾa$^{1}$**
CMPL.hit=1PL.EX=3.N=3SG.M
'We (ex.) hit him with it (the rock).'

Just as in SSZ, certain otherwise ungrammatical configurations are exceptionally permitted in this ditransitive construction. It is not surprising that 3rd person direct object clitics are licit in (36), given that they are already permitted in monotransitives. Instead, what is surprising is the fact that [PART] DO clitics are *also* permitted, regardless of whether the higher IO is also a clitic, (37a), or a lexical DP, (37b).

(37) a. kõ$^{13}$õ$^{3}$=ra$^{1}$ [ **ɲa$^{1}$ ʒu$^{1}$u$^{5}$ ʒo$^{5ʔ}$o$^{3}$** ]=**$^{n}$di$^{3}$**
CMPL.hit=3SG.M CL.3.N rock this =1PL.EX
'He hit us (ex.) with this rock.'

b. kõ$^{13}$õ$^{3}$=$^{n}$do$^{5}$=**ɲa$^{1}$=$^{n}$di$^{1}$**
CMPL.hit=2PL=3.N=1PL.EX
'You (pl.) hit us (ex.) with it (the rock).'

The SSZ and SJPM IO/DO combinations are summed up in (38) (differences between SSZ and SJPM are boxed). Given all of the other parallels demonstrated between SSZ and SJPM thus far, we take the examples in (37) to also illustrate the obviation of hierarchy effects. SJPM thus provides both further empirical grounding for this pattern and another key point of variation. As with other facts covered in this paper, we suggest that the pattern in SJPM ditransitives should inform any treatment of its SSZ counterpart (and vice versa).

(38) *SSZ and SJPM ditransitive internal arguments (where 1, 2, and 3 are clitics):*

| | SSZ | | | | SJPM | | | |
|---|---|---|---|---|---|---|---|---|
| IO | 3 | 3 | DP | DP | 3 | 3 | DP | DP |
| DO | 3 | 1/2 | 3 | 1/2 | 3 | 1/2 | 3 | 1/2 |
| | ✓ | $\boxed{✗}$ | ✓ | $\boxed{✗}$ | ✓ | $\boxed{✓}$ | ✓ | $\boxed{✓}$ |

That the strong PCC is lifted in SJPM (regardless of the IO's nominal properties) presents a challenge to the account of SSZ developed by Foley and Toosarvandani 2019. The SJPM pattern is actually *less* surprising than that in SSZ. If P's Probe Activation criterion is satisfied as soon as P encounters the subject and IO, then the Principle of Minimal Compliance dictates that [$\pi$] and [PART] direct objects alike should have no problem cliticizing. Therefore, it is the SSZ pattern that requires an additional stipulation: as noted above, Foley and Toosarvandani do not provide a concrete proposal, but do point to an independent Person Licensing Condition (PLC) requiring [PART] pronouns to be Agreed with (Béjar and Rezac, 2003). But this, in turn, becomes challenging to square with the SJPM facts, as we would have no explanation for why the PLC is lifted in SJPM ditransitives.

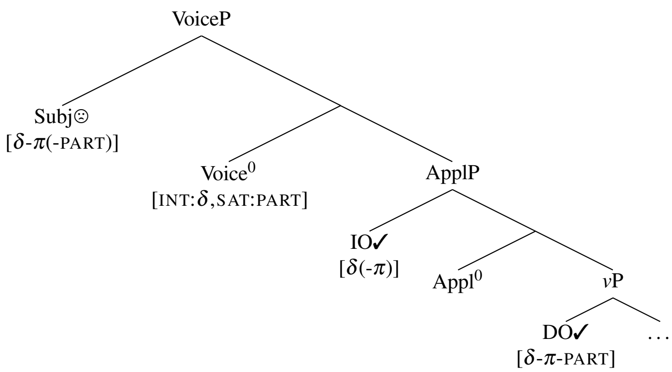
The analysis developed in §4, based on Deal 2024, also cannot readily account for the full range of ditransitive data across both languages. Recall that this analysis involves a probe P (bearing [INT:$\delta$,SAT:PART]) in Voice$^{0}$. This by itself fails to capture the grammaticality of [PART] DOs in SJPM ditransitives. In (39a),

compare SJPM with SSZ examples such as (i), which show that the strong PCC still holds between the higher two arguments.

(i) Ba ben=**ba'** $\boxed{\text{le'}}$ beku'
already CMPL.give=3.HU 2SG dog
'S/he gave you the dog.' (*2SG =*o'*) (Foley and Toosarvandani, 2019, p. 261)

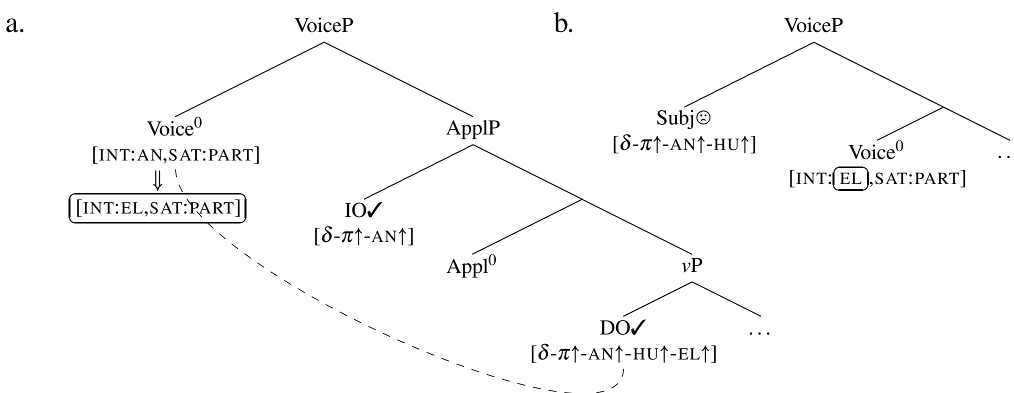
representing either example in (37), the probe in Voice$^0$ may interact with both the IO and DO in turn (for concreteness, the IO is in Spec-ApplP), but encountering the [PART] DO should halt the search process, preventing Agreement with the subject in Spec-VoiceP. In other words, this should yield the same Case Filter violation as in monotransitive LDB constructions.

(39)




By the same logic, this analysis is challenged by the SSZ example in (35b), in which the GCC is obviated. Recall from §4.1 that the GCC may be derived in an Int/Sat system if [$\pi$] and its animacy subfeatures are dynamic. For (35b), the probe's interaction condition is [INT:$\delta$], but gets updated to [INT:AN] and then [INT:EL] upon encountering the animate IO and the elder DO, respectively (only the latter is shown in (40a)). However, this should prevent the probe from interacting with the [HU] subject on its third cycle of Agree, since [EL] outranks [HU], (40b). Again, we expect a licensing violation here, contrary to fact.

(40)




We may try to resolve these issues via an additional probe in Appl$^0$ that targets the DO (in Appl$^0$'s c-command domain) and IO (in Spec-ApplP) in turn, coupled with a requirement such as the Activity Condition (Chomsky, 2000, 2001) rendering already-licensed arguments from being further Agreed with. The latter requirement would allow Voice$^0$ to Agree with its specifier, since the targets of Appl$^0$ would no longer be visible to Voice$^0$.[26] However, this overgenerates for SSZ. In particular, we would expect both SSZ and SJPM to permit DO clitics of *all persons* in ditransitives—but this is the case only in SJPM. In SSZ, recall that [PART] DOs are still prohibited, (35c). The problem is that the first argument targeted by Appl$^0$ is necessarily the DO, so Agree between the two should always be possible. A similar overgeneration issue arises for *DP/DP/3 configurations, which—like their *DP/3 monotransitive counterparts—are ungrammatical in SSZ, (41).

[26]Alternatively, we may specify ApplP as a phase (McGinnis, 2001), so that Voice$^0$ may never access the DO.

(41) *Blu'id=**b**$_i$ Maria bi'i nu'ule-dao $t_i$
CMPL.show=3.AN Maria CL girl-DIM
Intended: 'Maria showed it to the little girl.' (Foley and Toosarvandani, 2019, p. 259)

There may of course be other workarounds, though space considerations prevent us from developing these further. Regardless, we have suggested how neither account can *straightforwardly* accommodate the ditransitive patterns in both languages. More broadly, this discussion illustrates the utility of languages that display hierarchy restrictions simultaneously between different pairs of arguments in monotransitives and ditransitives. Such languages are not common, so most prior work on such effects have focused on one or the other (though see Stegovec 2019 and Johnson to appear). But we have highlighted two such attested patterns here as a fruitful avenue for further investigation.

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